MetricFlow

Google Ads API Tool Design Document

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1. Executive Summary

MetricFlow is a **read-only reporting and analytics platform** that helps businesses understand the relationship between their advertising spend and business outcomes. Users connect their Google Ads accounts (along with other platforms like Google Analytics, Facebook Ads, and QuickBooks) via OAuth, and MetricFlow syncs campaign performance metrics into a unified dashboard.

The platform provides cross-platform correlation analysis and AI-powered insights to help users understand which advertising campaigns are driving real business results.

> **Important:** MetricFlow is a reporting-only tool. It **never** creates, modifies, pauses, or deletes campaigns, ads, ad groups, keywords, or any other Google Ads entities. It only reads campaign performance data for display in dashboards and analytics.

2. Business Model

MetricFlow is a SaaS platform serving marketing teams and agencies that manage advertising campaigns across multiple platforms. The core value proposition is:

- **Unified Reporting:** View Google Ads, Facebook Ads, Google Analytics, and QuickBooks data in a single dashboard instead of switching between platforms.
- **Cross-Platform Correlation:** Automatically discover statistical relationships between advertising metrics (e.g., Google Ads spend) and business outcomes (e.g., revenue).
- **AI-Powered Insights:** Surface actionable recommendations based on data patterns that might otherwise go unnoticed.
- **Agency White-Label:** Agencies can brand the platform for their clients with custom logos and colors.

How We Use Google Ads

MetricFlow reads Google Ads campaign performance data via the Google Ads API to populate dashboards and run analytics. Specifically, we fetch daily campaign-level metrics (impressions, clicks, cost, conversions, CTR, CPC, and conversion value) and display them alongside data from other connected platforms. We sync data daily on an automated schedule and allow users to trigger manual syncs.

3. Tool Overview

Target Users

User Type	Description
Marketing Managers	Track campaign performance across platforms in one place
Business Owners	Understand how ad spend translates to revenue
Digital Agencies	Report to clients with white-labeled dashboards
Data Analysts	Explore cross-platform metric correlations

Core Features

Feature	Description	Google Ads Data Used
Platform Integrations	Connect Google Ads via OAuth; manage connected accounts	OAuth tokens, account metadata
Unified Dashboard	View all metrics in one place with time-series charts	Impressions, clicks, cost, conversions, CTR, CPC
Correlation Analysis	Discover statistical relationships between metrics	All synced Google Ads metrics correlated with other platforms
AI Insights	AI-generated recommendations based on data patterns	Correlation results involving Google Ads metrics
AI Chat	Natural language data exploration	All synced metrics queryable via chat
Sync History	View sync status and history for each integration	Sync timestamps, record counts, error status

4. Architecture & Data Flow

System Architecture

```
┌──────────────┐      OAuth 2.0      ┌──────────────────┐
│   User's     │◄────────────────────►│  Google OAuth    │
│   Browser    │                      │  Consent Screen  │
└──────┬───────┘                      └──────────────────┘
       │ HTTPS
       ▼
┌──────────────┐                      ┌──────────────────┐
│  MetricFlow  │   Google Ads API     │  Google Ads API  │
│  Web App     │──────────────────────►│  searchStream    │
│  (Phoenix)   │   (read-only)        │  (v23)           │
└──────┬───────┘                      └──────────────────┘
       │
       ▼
┌──────────────┐
│  PostgreSQL  │  Encrypted tokens + normalized metrics
│  Database    │
└──────────────┘
```

Data Flow: Google Ads Metric Sync

```
1. TRIGGER
   ├─ Daily automated sync (2:00 AM UTC via Oban cron)
   └─ Manual "Sync Now" button click by user

2. TOKEN CHECK
   ├─ Check if access_token is expired
   └─ If expired: refresh using refresh_token via Google OAuth

3. API REQUEST
   ├─ POST to googleads.googleapis.com/v23/customers/{id}/googleAds:searchStream
```

```
   ├─ Headers: Authorization (Bearer token), developer-token
   ├─ Body: GAQL query for campaign metrics, last 30 days
   └─ Response: JSON array of campaign performance rows

4. DATA PROCESSING
   ├─ Parse JSON response, extract metrics per campaign per day
   ├─ Normalize values (cost_micros ÷ 1,000,000 → dollars)
   └─ Transform to unified metric schema

5. STORAGE
   ├─ Insert normalized metrics into PostgreSQL
   └─ Record sync history (timestamp, count, status)

6. DISPLAY
   ├─ Dashboard queries metrics for visualization
   ├─ Correlation engine analyzes cross-platform relationships
   └─ AI generates insights from correlation results
```

5. Google Ads API Usage

API Endpoints Used

Endpoint	Method	Purpose	Frequency
`customers/{id}/googleAds:searchStream`	POST	Fetch campaign performance metrics	Once daily + manual triggers

> MetricFlow uses **only one API endpoint** — the searchStream endpoint for reading campaign performance data. No campaign management, creation, or modification endpoints are used.

GAQL Query

The following GAQL query is used to fetch campaign metrics:

```
SELECT
  campaign.id,
  campaign.name,
  campaign.status,
  segments.date,
  metrics.impressions,
  metrics.clicks,
  metrics.cost_micros,
  metrics.conversions,
  metrics.ctr,
  metrics.average_cpc,
  metrics.conversions_value
FROM campaign
WHERE segments.date BETWEEN '{start_date}' AND '{end_date}'
  AND campaign.status != 'REMOVED'
ORDER BY segments.date DESC
```

Metrics Retrieved

Metric	Type	Display Name	Conversion
metrics.impressions	Integer	Impressions	None

metrics.clicks	Integer	Clicks	None
metrics.cost_micros	Integer (micros)	Spend	÷ 1,000,000 → dollars
metrics.conversions	Float	Conversions	None
metrics.ctr	Float (0–1)	CTR	× 100 → percentage
metrics.average_cpc	Integer (micros)	Avg CPC	÷ 1,000,000 → dollars
metrics.conversions_value	Float	Conv. Value	None

Authentication Headers

Header	Value	Source
`Authorization`	`Bearer {user_access_token}`	Per-user OAuth token (encrypted in DB)
`developer-token`	`{developer_token}`	Application-level env var
`login-customer-id`	`{mcc_id}` (optional)	For MCC/manager accounts

Rate Limiting & Error Handling

Scenario	HTTP Status	MetricFlow Behavior
Success	200	Parse and store metrics
Token expired	401	Refresh token, retry once
Permission denied	403	Log error, notify user
Rate limited	429	Retry with exponential backoff (up to 3 attempts)
Server error	500+	Retry with backoff, mark sync as failed

6. User Interface — Screen-by-Screen Walkthrough

6.1 Home Page (Public)

The landing page at `metric-flow.app` describes the platform's features and provides links to register, sign in, and view the privacy policy and terms of service.

Elements:

- **"Get Started" button:** Navigates to user registration page
- **"Sign In" button:** Navigates to login page
- **Feature cards:** Describe Platform Integrations, Unified Dashboards, Correlation Analysis, AI Insights, AI Chat, and Teams & White Label
- **Footer:** Links to Privacy Policy, Terms of Service, and Contact email

6.2 Integration Connection Screen

Route: /integrations/connect

```
┌──────────────────────────────────────────────────────────────┐
│  [MetricFlow Logo]   Integrations  Correlations  ...  [User]│
├──────────────────────────────────────────────────────────────┤
│                                                              │
│  Connect a Platform                                          │
│                                                              │
│  ┌──────────────┐  ┌──────────────┐  ┌──────────────┐       │
│  │  Google Ads  │  │  Google      │  │  Facebook    │       │
│  │  [G icon]    │  │  Analytics   │  │  Ads         │       │
│  │  [Connect]   │  │  [Connect]   │  │  [Connect]   │       │
│  └──────────────┘  └──────────────┘  └──────────────┘       │
│                                                              │
│  ┌──────────────┐  ┌──────────────┐                         │
│  │  QuickBooks  │  │  GitHub      │                         │
│  │  [QB icon]   │  │  [GH icon]   │                         │
│  │  [Connect]   │  │  [Connect]   │                         │
│  └──────────────┘  └──────────────┘                         │
│                                                              │
└──────────────────────────────────────────────────────────────┘
```

Elements:

- **"Connect" button (Google Ads):** Initiates OAuth 2.0 flow → redirects to Google consent screen requesting `adwords` scope with `access_type=offline` and `prompt=consent`
- After authorization, user is redirected back to MetricFlow with an authorization code
- MetricFlow exchanges the code for access + refresh tokens and stores them encrypted

6.3 Integration Management Screen

Route: /integrations

```
┌──────────────────────────────────────────────────────────────┐
│  [MetricFlow Logo]   Integrations  Correlations  ...  [User]│
├──────────────────────────────────────────────────────────────┤
│                                                              │
│  Your Integrations              [+ Connect Platform]         │
│                                                              │
│  ┌──────────────────────────────────────────────────────┐   │
│  │  Google Ads          ● Connected   Last sync: 2h ago │   │
│  │  Customer ID: 123-456-7890                           │   │
│  │  [Sync Now]  [Edit Accounts]  [Disconnect]           │   │
│  └──────────────────────────────────────────────────────┘   │
│                                                              │
│  ┌──────────────────────────────────────────────────────┐   │
│  │  Google Analytics    ● Connected   Last sync: 2h ago │   │
│  │  Property: UA-12345                                  │   │
│  │  [Sync Now]  [Edit Accounts]  [Disconnect]           │   │
│  └──────────────────────────────────────────────────────┘   │
│                                                              │
└──────────────────────────────────────────────────────────────┘
```

Elements:

- **"Sync Now" button:** Enqueues an immediate data sync job for this integration. Fetches the last 30 days of campaign metrics from Google Ads API and stores them in the database.
- **"Edit Accounts" link:** Opens account selection to choose which Google Ads customer accounts to sync.
- **"Disconnect" button:** Revokes OAuth tokens and removes the integration. All synced metrics remain in the database until the user deletes their account.
- **Status badge:** Shows "Connected" (green) or "Error" (red) based on last sync result.

6.4 Unified Dashboard

Route: /dashboard

```
┌──────────────────────────────────────────────────────────────┐
│  [MetricFlow Logo]   Integrations  Correlations  ...  [User]│
├──────────────────────────────────────────────────────────────┤
│                                                              │
│  Dashboard                    [Date Range ▼] [Platform ▼]   │
│                                                              │
│  ┌──────────┐ ┌──────────┐ ┌──────────┐ ┌──────────┐       │
│  │ Clicks   │ │ Spend    │ │ Impress. │ │ Convers. │       │
│  │ 12,450   │ │ $4,230   │ │ 450K     │ │ 320      │       │
│  │ +12%     │ │ -5%      │ │ +8%      │ │ +15%     │       │
│  └──────────┘ └──────────┘ └──────────┘ └──────────┘       │
│                                                              │
│  ┌──────────────────────────────────────────────────────┐   │
│  │              Time Series Chart (Vega-Lite)           │   │
│  │     /\                                               │   │
│  │    /  \    /\  /\                                    │   │
│  │   /    \  /  \/  \                                   │   │
│  │  /      \/        \___                               │   │
│  │  Mar 1   Mar 8   Mar 15   Mar 22   Mar 29            │   │
│  └──────────────────────────────────────────────────────┘   │
│                                                              │
│  Derived Metrics                                             │
│  ┌──────────┐ ┌──────────┐ ┌──────────┐                    │
│  │ CPC      │ │ CTR      │ │ ROAS     │                    │
│  │ $0.34    │ │ 2.8%     │ │ 3.2x     │                    │
│  └──────────┘ └──────────┘ └──────────┘                    │
│                                                              │
└──────────────────────────────────────────────────────────────┘
```

Elements:

- **Summary stat cards:** Display aggregated metrics from all connected platforms (or filtered to Google Ads). Raw metrics: Clicks, Spend (from cost_micros), Impressions, Conversions. Derived metrics: CPC (Spend/Clicks), CTR (Clicks/Impressions), ROAS (Revenue/Spend).
- **Time series chart:** Interactive Vega-Lite visualization showing metrics over time, broken down by day.
- **Date range selector:** Filter data by last 7, 14, 30, or 90 days.
- **Platform filter:** Show all platforms or filter to just Google Ads data.
- **All data is read-only.** No buttons or features modify Google Ads campaigns.

6.5 Correlation Analysis

Route: /correlations

```
┌──────────────────────────────────────────────────────────────┐
│  [MetricFlow Logo]   Integrations  Correlations  ...  [User]│
├──────────────────────────────────────────────────────────────┤
│                                                              │
│  Correlation Analysis                          [Run Now]     │
│                                                              │
│  ┌──────────────────────┬────────┬──────┬──────────┬────────┐ │
│  │ Metric Pair          │ Corr.  │ Lag  │ Strength │ Window │ │
│  ├──────────────────────┼────────┼──────┼──────────┼────────┤ │
│  │ Google Ads Spend     │        │      │          │        │ │
│  │ vs. QB Revenue       │ 0.92   │ 3d   │ Strong   │ 30d    │ │
│  ├──────────────────────┼────────┼──────┼──────────┼────────┤ │
│  │ Google Ads Clicks    │        │      │          │        │ │
│  │ vs. Conversions      │ 0.78   │ 0d   │ Strong   │ 30d    │ │
│  ├──────────────────────┼────────┼──────┼──────────┼────────┤ │
│  │ Google Ads Cost      │        │      │          │        │ │
│  │ vs. FB Ad Spend      │ 0.45   │ 1d   │ Moderate │ 30d    │ │
│  └──────────────────────┴────────┴──────┴──────────┴────────┘ │
│                                                              │
│  [Switch to Smart Mode — AI-powered recommendations]         │
│                                                              │
└──────────────────────────────────────────────────────────────┘
```

Elements:

- **"Run Now" button:** Triggers a new correlation analysis job. Computes Pearson correlation coefficients between all pairs of synced metrics (including Google Ads metrics paired with other platform metrics).
- **Results table:** Shows metric pairs ranked by correlation strength, with optimal time lag and data window.
- **"Smart Mode" toggle:** Switches from raw correlation table to AI-generated natural language recommendations.
- **All data is read-only.** Correlation analysis only reads stored metrics.

6.6 AI Insights

Route: /insights

```
┌──────────────────────────────────────────────────────────────┐
│  [MetricFlow Logo]   Integrations  Correlations  ...  [User]│
```

```
├──────────────────────────────────────────────────────────┤
│                                                          │
│  AI Insights                                             │
│                                                          │
│   ┌──────────────────────────────────────────────────┐   │
│   │  💡 Budget Optimization            Confidence: 85% │   │
│   │                                                  │   │
│   │  Your Google Ads spend shows a 0.92 correlation  │   │
│   │  with QuickBooks revenue with a 3-day lag. Consider │   │
│   │  increasing your Google Ads budget as it appears │   │
│   │  to be a strong driver of business revenue.      │   │
│   │                                                  │   │
│   │  Metrics: Google Ads Cost ↔ Revenue              │   │
│   │  [👍 Helpful]  [👎 Not Helpful]                  │   │
│   └──────────────────────────────────────────────────┘   │
│                                                          │
│   ┌──────────────────────────────────────────────────┐   │
│   │  📊 CTR Monitoring                 Confidence: 62% │   │
│   │                                                  │   │
│   │  Google Ads CTR shows weak correlation with      │   │
│   │  conversions. This may indicate that click quality │   │
│   │  needs improvement — consider refining ad targeting. │   │
│   │                                                  │   │
│   │  [👍 Helpful]  [👎 Not Helpful]                  │   │
│   └──────────────────────────────────────────────────┘   │
│                                                          │
└──────────────────────────────────────────────────────────┘
```

Elements:

- **Insight cards:** AI-generated recommendations based on correlation analysis results. Each card shows: insight type (budget, optimization, monitoring), confidence score, natural language explanation, and the metrics involved.
- **Feedback buttons:** "Helpful" / "Not Helpful" — used to improve future insight quality.
- **All insights are informational.** No buttons or features modify Google Ads campaigns. Users must go to Google Ads directly to act on recommendations.

6.7 Sync History

Route: /integrations/sync-history

```
┌──────────────────────────────────────────────────────────┐
│  [MetricFlow Logo]   Integrations  Correlations  ...  [User]│
├──────────────────────────────────────────────────────────┤
│                                                          │
```

```
│  Sync History                                                  │
│                                                                 │
│  ┌──────────────┬────────────┬────────┬──────────────────┐ │
│  │ Provider     │ Timestamp  │ Status │ Records Synced   │ │
│  ├──────────────┼────────────┼────────┼──────────────────┤ │
│  │ Google Ads   │ Mar 9, 2AM │ ✓ OK   │ 2,140 metrics    │ │
│  │ Google Ads   │ Mar 8, 2AM │ ✓ OK   │ 2,130 metrics    │ │
│  │ Google Ads   │ Mar 7, 2AM │ ✗ Fail │ Token expired    │ │
│  │ Facebook Ads │ Mar 9, 2AM │ ✓ OK   │ 890 metrics      │ │
│  └──────────────┴────────────┴────────┴──────────────────┘ │
│                                                                 │
└─────────────────────────────────────────────────────────────────┘
```

7. OAuth Authorization Flow

Flow Diagram

```
1. User clicks "Connect" on Google Ads card
       |
2. MetricFlow redirects to Google OAuth consent screen
   URL: accounts.google.com/o/oauth2/auth
   Parameters:
     scope        = https://www.googleapis.com/auth/adwords
                    email profile
     access_type  = offline    (request refresh token)
     prompt       = consent    (force consent to ensure refresh token)
     redirect_uri = https://metric-flow.app/integrations/oauth/callback/google
       |
3. User reviews permissions and clicks "Allow"
       |
4. Google redirects to MetricFlow callback with authorization code
       |
5. MetricFlow exchanges code for tokens via Google token endpoint
   Response: { access_token, refresh_token, expires_in }
       |
6. Tokens encrypted (AES-256-GCM) and stored in PostgreSQL
   access_token  → encrypted in integrations.access_token
   refresh_token → encrypted in integrations.refresh_token
   expires_at    → calculated from expires_in
       |
7. User redirected to /integrations — sees "Connected" status
```

OAuth Scopes Requested

Scope	Purpose
`https://www.googleapis.com/auth/adwords`	Read Google Ads campaign performance data
`email`	Identify user's Google account

`profile`	Display user's name in the UI

Token Refresh

When an access token expires, MetricFlow automatically refreshes it using the stored refresh token before making API requests. If the refresh fails (e.g., user revoked access), the integration is marked with an error status and the user is prompted to re-authorize.

8. Data Storage & Security

Encryption

Data	Storage	Encryption
OAuth access tokens	PostgreSQL	AES-256-GCM at rest (Cloak library)
OAuth refresh tokens	PostgreSQL	AES-256-GCM at rest (Cloak library)
User passwords	PostgreSQL	bcrypt hash (never stored in plaintext)
Synced metrics	PostgreSQL	Standard DB encryption, TLS in transit

Infrastructure Security

- **Transport:** All connections use TLS/SSL (HTTPS enforced)
- **Hosting:** Hetzner Cloud (Germany), Docker containers with isolated networks
- **Database:** PostgreSQL 17, not exposed to public internet, internal Docker network only
- **Secrets:** Environment variables, never committed to source code
- **CSP Headers:** Content Security Policy restricts page resources to same origin

Access Controls

- All API data is scoped to the authenticated user — users can only see their own Google Ads data
- OAuth tokens are per-user and never shared between accounts
- Agency access levels are configurable by account administrators
- Unique constraint enforces one Google Ads integration per user

9. Data Retention & Deletion

- **Active account:** Synced metrics retained as long as the account is active
- **Disconnected integration:** OAuth tokens immediately revoked and deleted; synced metrics remain for historical analysis
- **Deleted account:** All data (tokens, metrics, sync history) permanently removed within 30 days
- **Data deletion requests:** Users can request deletion at any time via privacy@metric-flow.app

10. Privacy & Compliance

MetricFlow's use and transfer to any other app of information received from Google APIs adheres to the Google API Services User Data Policy, including the Limited Use requirements.

- We only access Google data explicitly authorized through the OAuth consent flow
- We use Google data solely to provide MetricFlow's reporting and analytics features
- We do not sell Google user data to third parties
- We do not use Google user data for advertising purposes
- We do not allow humans to read Google user data unless required for security, legal compliance, or with explicit user consent

Full privacy policy: https://metric-flow.app/privacy
Terms of service: https://metric-flow.app/terms